

Project Pitch: Indoor Cleaning Robot (Voom-Bot)

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Target Specialisation: Option B - Autonomous Robotics (Webots)

Hardware Platform: Pioneer 3-DX

Executive Summary

Voom-Bot is an autonomous mobile robot designed to navigate indoor environments such as classrooms, libraries and living spaces while performing routine cleaning. The project focuses on perception driven navigation, where the robot explores an unfamiliar environment, the system uses a finite state machine with recovery, cliff avoidance, charging, cleaning, exploration and obstacle avoidance states.

The robot operates fully autonomously using a combination of sensor inputs and structured decision making to ensure both efficiency and safety for users and surrounding objects.

System Architecture (Sense–Think–Act)

The Pioneer 3-DX model in Webots provides a stable and realistic platform for indoor navigation and control.

- **Sensing (Inputs):**
 1. **Distance Sensors (Sonar/Proximity):** Multiple sensors provide 360° proximity data, enabling reliable obstacle detection and collision avoidance.
 2. **Cliff Distance Sensors (Sonar/Proximity):** Attached to the bottom front left and right of the robot for cliff detection.
 3. **GPS:** Provides global position for mapping and navigation.
 4. **Compass:** Provides orientation for heading correction.
 5. **Keyboard Input:** To simulate user interaction with the device.
- **Processing (Thinking):**
 1. **Finite State Machine (Core Control Logic):** This is the brain that controls different behaviours, allowing clear, structured design and analysis. It makes complex decisions to prevent collisions and prioritise safety.
 2. **Reactive Obstacle Perception Module:** The Perception Module uses reactive, event-driven logic where the robot continuously monitors the environment and triggers a predefined behavioural response whenever a specific condition is detected, such as an obstacle completed task.
 3. **Multi-Condition Decision Logic:** The system will use multi-condition decision logic by combining sensor inputs (distance, vision, audio) with internal states such as battery level to better determine behaviour transitions. Safety conditions override all other behaviours.
 4. **Grid Based Coverage Mapping:** Tracks visited cells and dynamically assigns cleaning targets using structured grid traversal.
 5. **Fault Detection System:** Includes stuck detection and cliff detection for safety overrides.
- **Actuation (Outputs):**
 1. **LED Indicators:** Provide visual feedback for robot states (cleaning, paused, error).
 2. **Drive Motors:** Use wheel rotations to allow movement, which allows the robot's navigation.
 3. **Speaker:** To notify users that the robot is incapable of operating (Unsafe or Out of power).
 4. **Vacuum Attachment (Vacuum Placeholder):** Simulates the cleaning function while the robot is in cleaning state.

Behavioural Logic (FSM)

State	Trigger Condition	Action / Output
PAUSED	Startup / manual toggle	Motors stopped, system idle
EXPLORE	Default active mode	Wall following exploration using distance sensors
CLEANING	Exploration complete + coverage mode	Grid-based target navigation cleaning
AVOID	Obstacle detected	Reactive turning based on sensor weighting
CLIFF	Drop detected by cliff sensors	Reverse + rotate away from edge
RECOVERY	Robot stuck detected	Stop motors, recovery behavior + alerts
CHARGING	Low battery or charge command	Navigate to dock and recharge

Specialisation: Autonomous Robotics

To meet the **Option B** requirements, Voom-Bot implements:

- **Obstacle Avoidance:** It uses Distance Sensors with set distance thresholds to avoid collisions with objects in its proximity.
- **Perception-Driven Decision:** It analyses data from the Distance Sensors, and human input to plan an efficient cleaning path.
- **Navigation:** Exploration based navigation using grid mapping to track visited vs unvisited areas and improve coverage efficiency.

Safety & Robustness

- **Human/Pet Detection:** Detects nearby movement or objects using sensors and pauses operation to prevent collision.
- **Low Battery Protection:** Triggers return to dock when battery falls below a threshold.
- **Cliff/Drop Detection:** Distance sensors and the camera detect ledges, like stairs, and stops the robot from falling.
- **Stuck Detection:** If the wheels spin without motion, the robot is set to recovery to prevent motor damage.

Advanced Component (10%)

Autonomous Navigation (A Grid-Based Mapping): Voom-Bot implements a grid based coverage system that tracks visited cells in real time, assigns cleaning targets using zigzag traversal, computes coverage percentage of the environment, automatically transitions between exploration and cleaning modes, and detects completion and triggers return to dock.